Automated Test Project

Project Requirements & Architectural Specification Blueprint

1. Executive Summary

Automated Test Project is designed as an intelligent, automated digital platform specifically tailored to solve manual workflow friction for Internal Staff & Academic Faculty. The platform centralizes data operations, eliminates redundant paperwork, and enforces data accuracy and compliance with sub-second responsiveness.

2. Problem Statement

Currently, stakeholders rely on disjointed spreadsheets, manual data compilation, and error-prone physical paperwork. This leads to operational bottlenecks, delayed reports, and an estimated 10-15 wasted hours per staff member every week.

3. Project Goals

- Automate end-to-end data processing for Automated Test Project
- Enable high-velocity document export in compliant PDF, Word, and Excel formats
- Provide auditable, role-based security ensuring sensitive student and organizational records remain protected
- Reduce report compilation latency from days to under 10 seconds

4. Confirmed Requirements

Requirement	Category	Priority	MVP
Role-Based Authentication & Session Management	Security	Must Have	Yes
Automated Document Rendering Engine	Functional	Must Have	Yes
Spreadsheet Ingestion & Validation Pipeline	Data Management	Must Have	Yes
Interactive Analytics & Insights Dashboard	Reporting	Should Have	No

5. Page-by-Page UI Architecture

Login & Onboarding — *Route: /login*

Key Components: Email/password form, SSO button, Password recovery, Role preview

Executive Dashboard — *Route: /dashboard*

Key Components: Recent activities, quick action launchpad, system status cards, pending approvals

Document & Academic Workspace — *Route: /workspace*

Key Components: File upload dropzone, live template previewer, export toolbar (PDF/DOCX/Excel)

Analytics & Results Center — *Route: /analytics*

Key Components: Interactive charts, pass percentage distribution, arrear identification table

Administration & Settings — *Route: /admin*

Key Components: User role management, question templates editor, security & audit trail viewer

6. Security, Compliance & Governance

- Password hashing using industry-standard bcrypt with salt
- JWT tokens signed with HS256 and configurable expiration
- Server-side authorization guards preventing unauthorized tenant access
- Sanitization of all incoming filenames and uploaded document streams
- Audit trail tracking all export and project administrative operations

7. Acceptance Criteria

- [ ] User can input a project idea in English or Tamil-English transliteration and have it auto-structured
- [ ] Questions adapt in real time to previous selections without page reloads
- [ ] Irrelevant questions are automatically omitted from the active flow
- [ ] Auto-save preserves every answer with visual confirmation
- [ ] Clicking export generates real downloadable PDF, DOCX, Markdown, JSON, and CSV files
- [ ] The entire platform runs responsively across mobile and desktop viewports